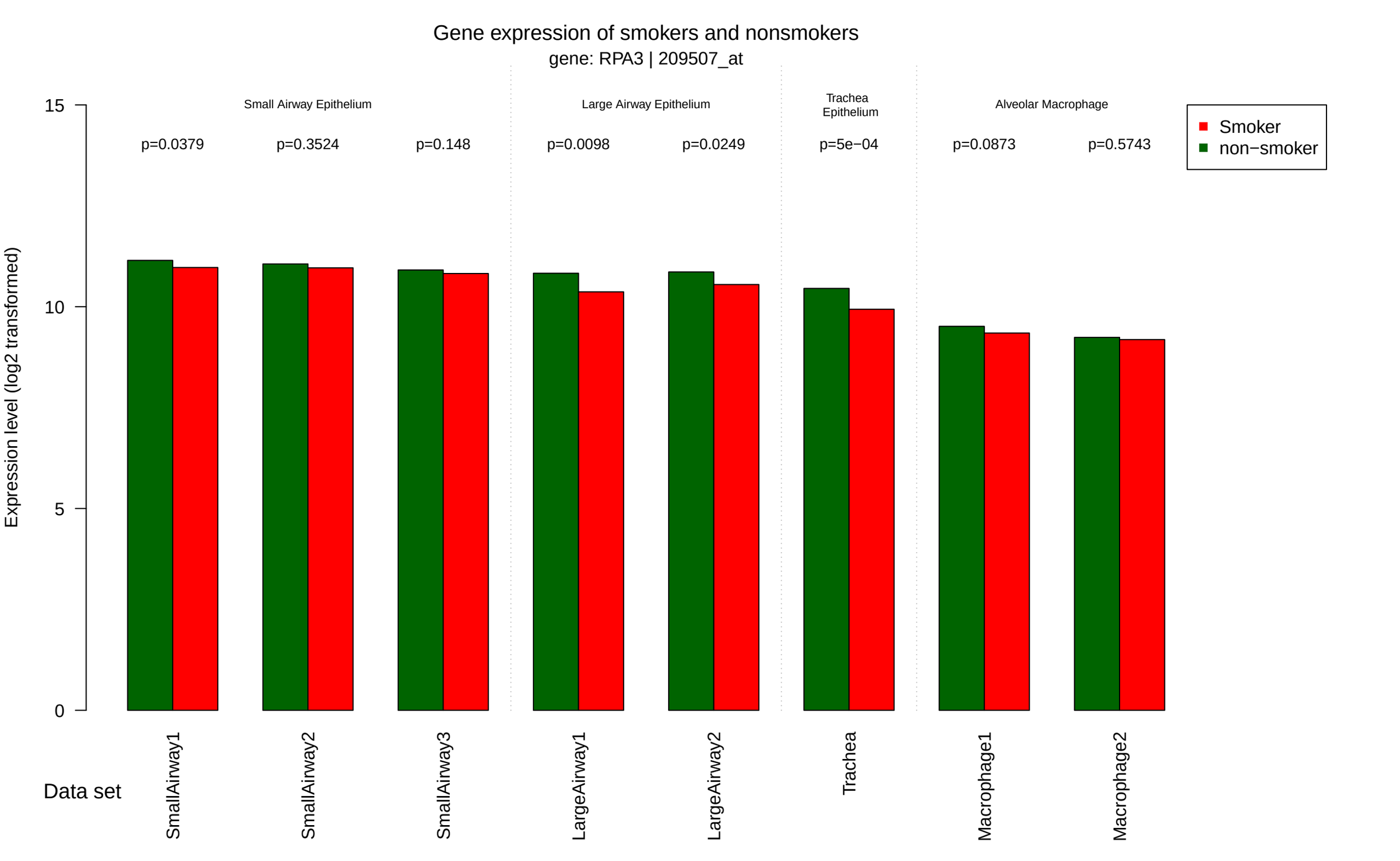
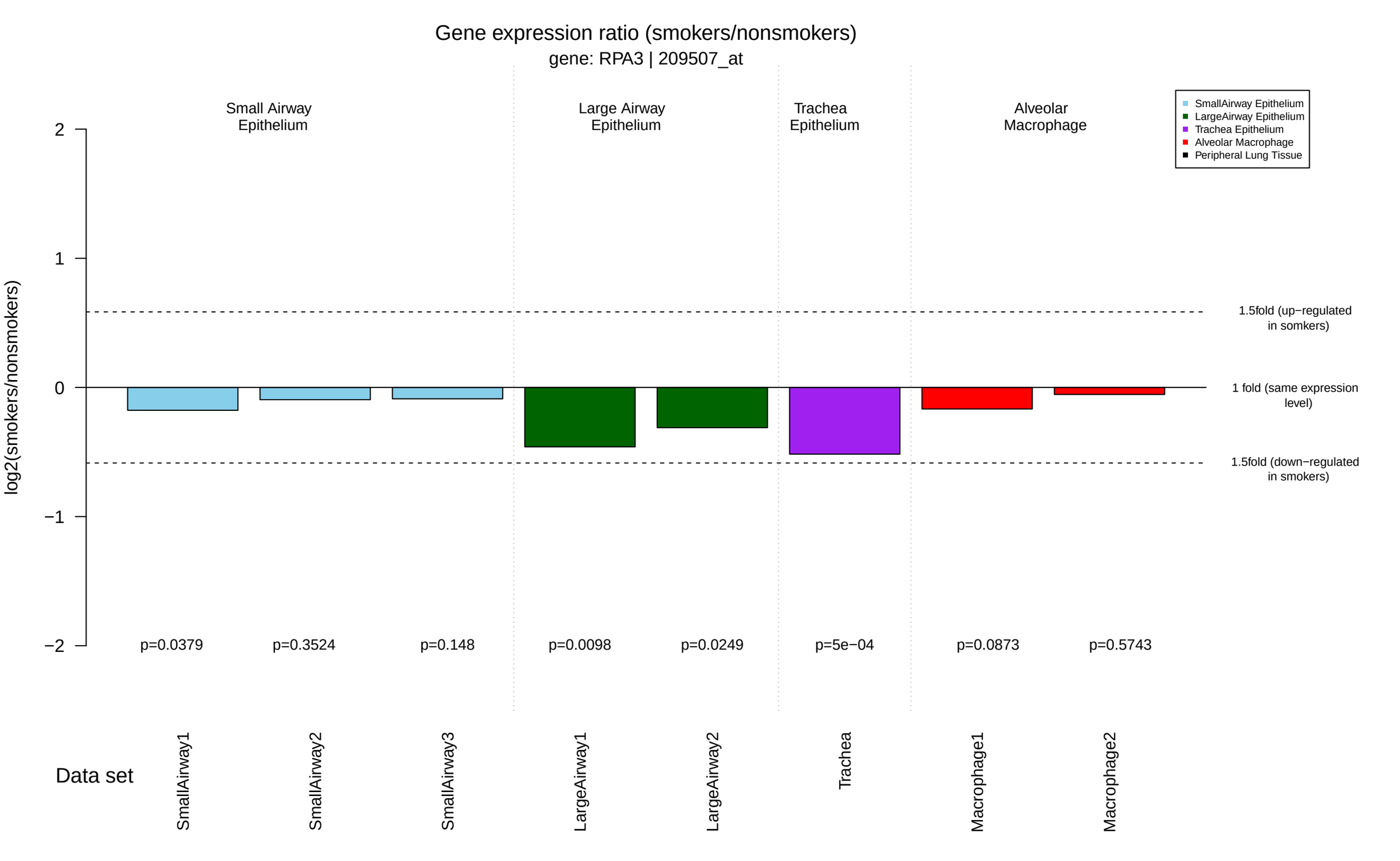
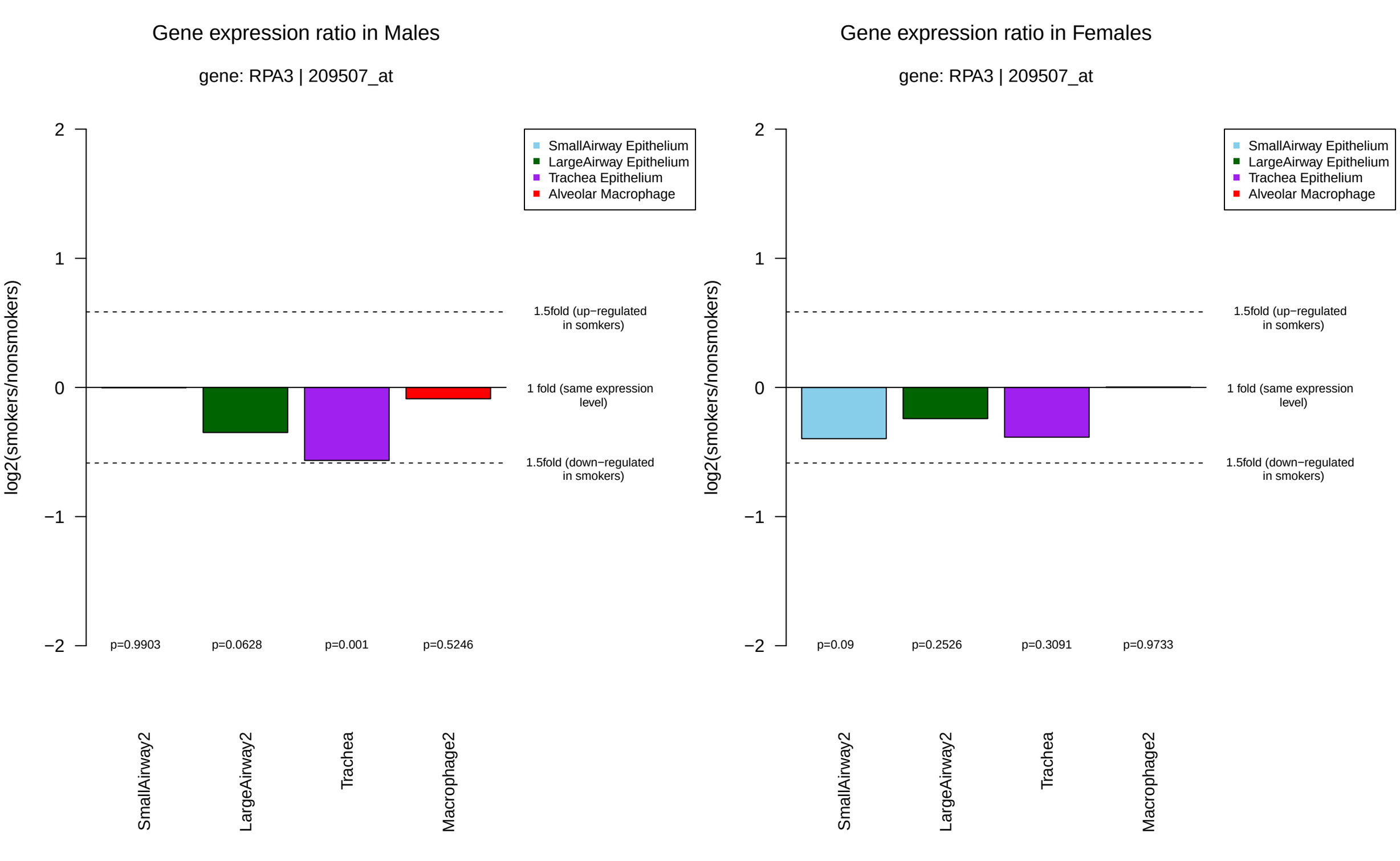
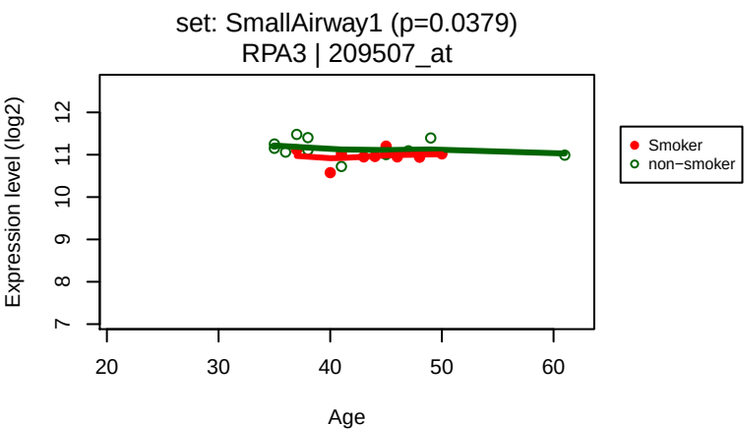
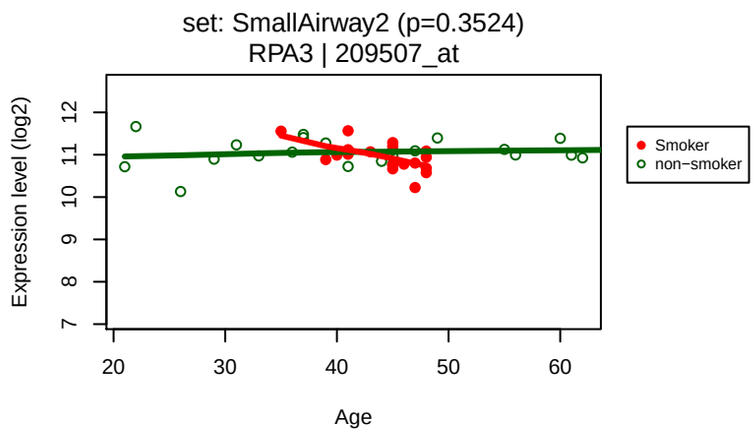
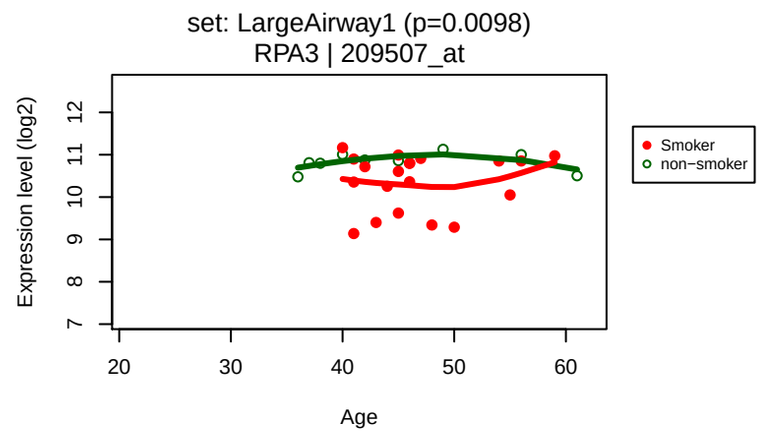

Gene expression of smokers and nonsmokers
gene: RPA3 | 209507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0379
p=0.3524
p=0.148
p=0.0098
p=0.0249
p=5e−04
p=0.0873
p=0.5743
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

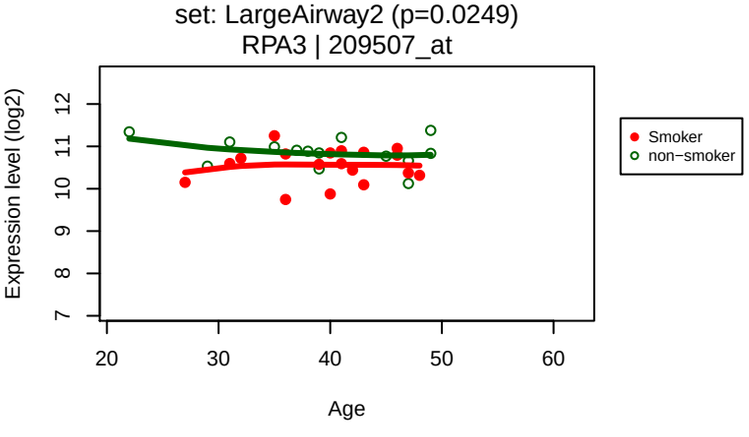

Gene expression ratio (smokers/nonsmokers)
gene: RPA3 | 209507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0379
p=0.3524
p=0.148
p=0.0098
p=0.0249
p=5e−04
p=0.0873
p=0.5743
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

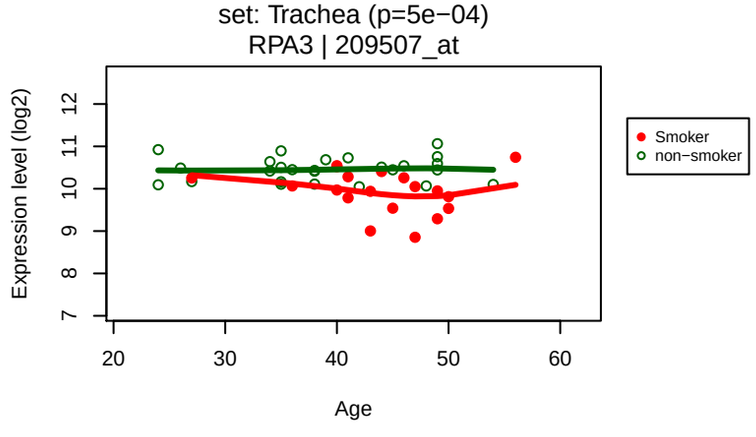

Gene expression ratio in Males
gene: RPA3 | 209507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9903
p=0.0628
p=0.001
p=0.5246
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPA3 | 209507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.09
p=0.2526
p=0.3091
p=0.9733
SmallAirway2
LargeAirway2
Trachea
Macrophage2

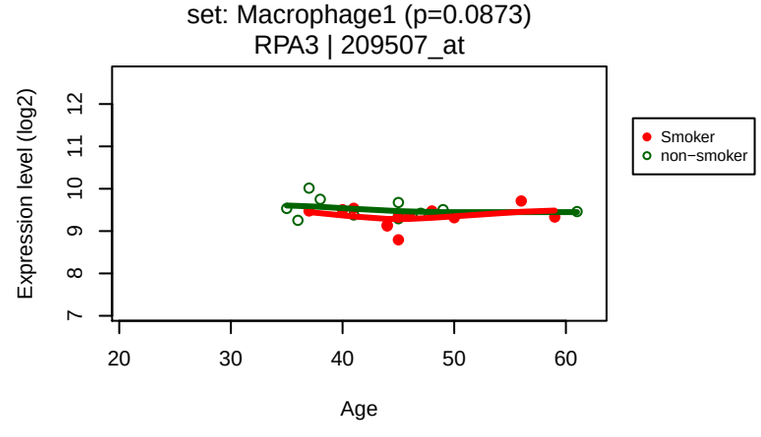

set: SmallAirway1 (p=0.0379)
RPA3 | 209507_at
Expression level (log2)
Age
Smoker
non−smoker

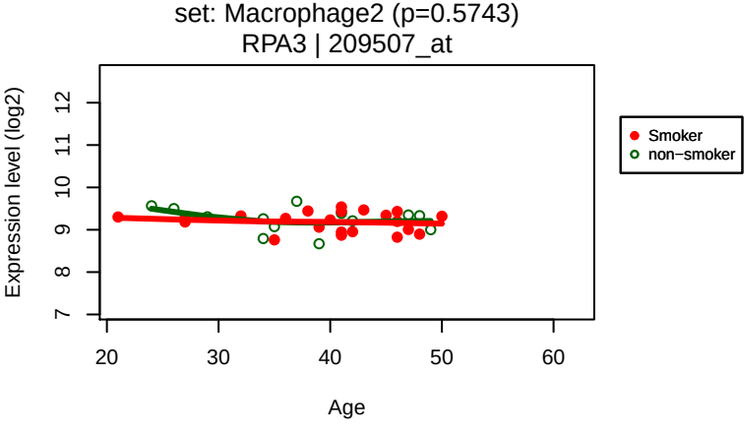

set: SmallAirway2 (p=0.3524)
RPA3 | 209507_at
Expression level (log2)
Age
Smoker
non−smoker

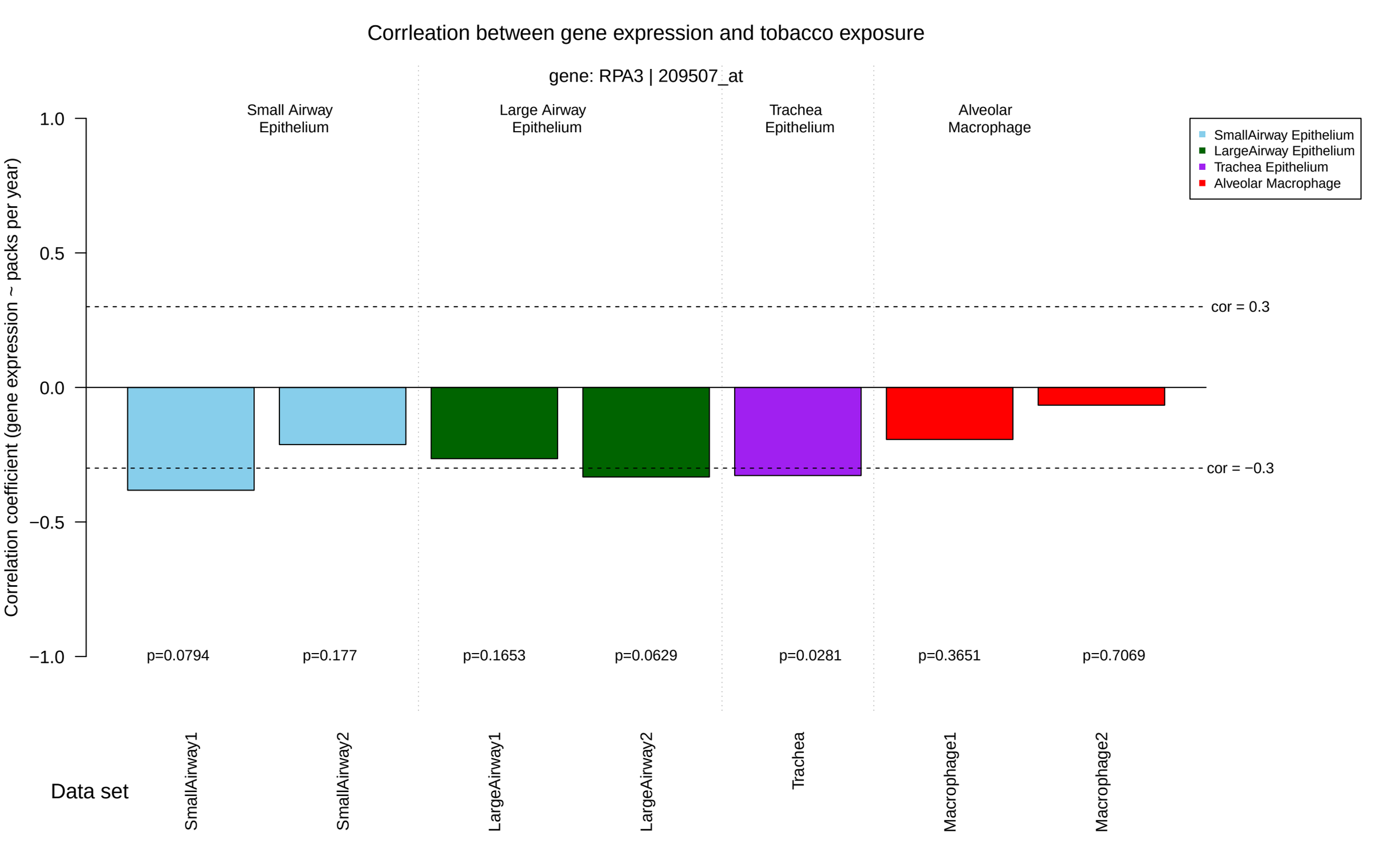

set: LargeAirway1 (p=0.0098)
RPA3 | 209507_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0249)
RPA3 | 209507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
RPA3 | 209507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0873)
RPA3 | 209507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5743)
RPA3 | 209507_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPA3 | 209507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0794
p=0.177
p=0.1653
p=0.0629
p=0.0281
p=0.3651
p=0.7069
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2